

Workshop 1

Problem 1.

Keywords: Intervals, Absolute Value, Domain

Inequalities are used frequently to describe subsets of real numbers or to give constraints in applications. We need to be able to solve the inequalities used and find useful ways in which to describe the solutions to them. In this problem we will consider some exercises that involve inequalities.

- (a) Students in a first semester calculus course were given a diagnostic quiz to see how much algebra and precalculus they remembered from the prerequisite courses. One of the problems asked students to solve the inequality

$$x^2 - 2x - 4 > -2x + 5$$

Here is a solution one of the students submitted:

$$\begin{array}{rclcl} x^2 - 2x - 4 & > & -2x + 5 \\ x^2 & > & 9 & \implies & \text{The answer is } x > \pm 3. \\ x & > & \pm 3 \end{array}$$

The above solution is incorrect. Write out an explanation using 1-2 English sentences of why the above solution is not valid. And now that you understand why the given solution is not correct, write a correct solution.

- (b) Consider the interval $[-3, 5]$. Find the length and middle point of the interval. Express the interval in terms of an inequality involving an absolute value.

- (c) Find the domain of $f(x) = \frac{4x + 5}{\sqrt{3x^2 - 10x - 8}}$. Express the domain using interval notation and then express your interval in terms of an inequality involving absolute value. Report your final answer in a brief sentence. For example, if your domain is given by the interval was $[2, 4]$, you would write:

The domain is the interval $[2, 4]$. The interval $[2, 4]$ can be described using the inequality $|x - 3| \leq 1$.

Problem 2.

Keywords: Quadratic Functions, Variable Parameters

In this problem, we will explore quadratic functions and their roots.

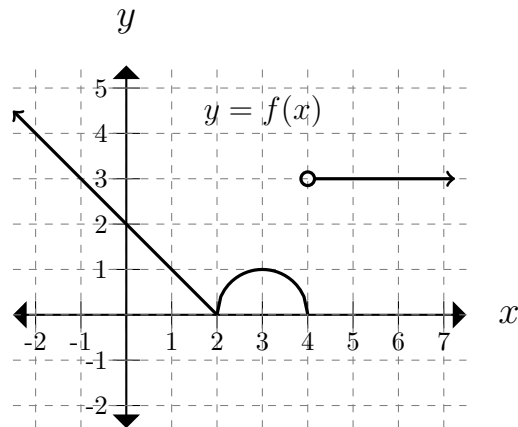
- (a) Consider the quadratic function $f(x) = x^2 + 4x + 3$. Find the vertex of f , the roots of f (if it has any) and sketch a graph of f .
- (b) Using your answer to part (a), determine **without making any computation** whether the function $f(x) = x^2 + 4x + 8$ has roots. Explain your answer carefully.
- (c) Consider the function $f(x) = x^2 + 4x + c$. For what values of c does this functions have no roots? A double root? Two distinct roots?

- (d) Why do your answers to part (c) make sense? Consider the graph of the function $g(x) = x^2 + 4x$. What are we doing to the graph of this function by changing c ?
- (e) Now, consider the function $f(x) = x^2 + bx + 4$. For what values of b do we have no roots? A double root? Two distinct roots?

Problem 3.

Keywords: Piecewise-defined functions, Lines, Circles, Shifting, Reflections

Let $f(x)$ be the function whose graph is given below:



- (a) Write $f(x)$ as a piecewise-defined function.

$$f(x) = \begin{cases} \boxed{} & \text{if } x < 2 \\ \boxed{} & \text{if } 2 \leq x \leq 4 \\ \boxed{} & \text{if } \boxed{} \end{cases}$$

- (b) Sketch the graph of $y = -f(x+2) + 3$ on the set of axes below. Be sure to label the axes and any points of interest.

